Assembly Scrap Intelligence

Multi-Factor Defect Prediction via XGBoost

"When twelve variables interact to produce a defect, you need a model that thinks about all twelve simultaneously."

0.593	69%	10,000	200
ROC-AUC	Recall @ thr=0.25	Assembly Cycles	Boosting Trees

XGBoost	Binary Classification	SHAP Explainability	Threshold Optimisation

Electronics & Mechatronics Assembly · Scrap & Quality Control · Shift Analysis

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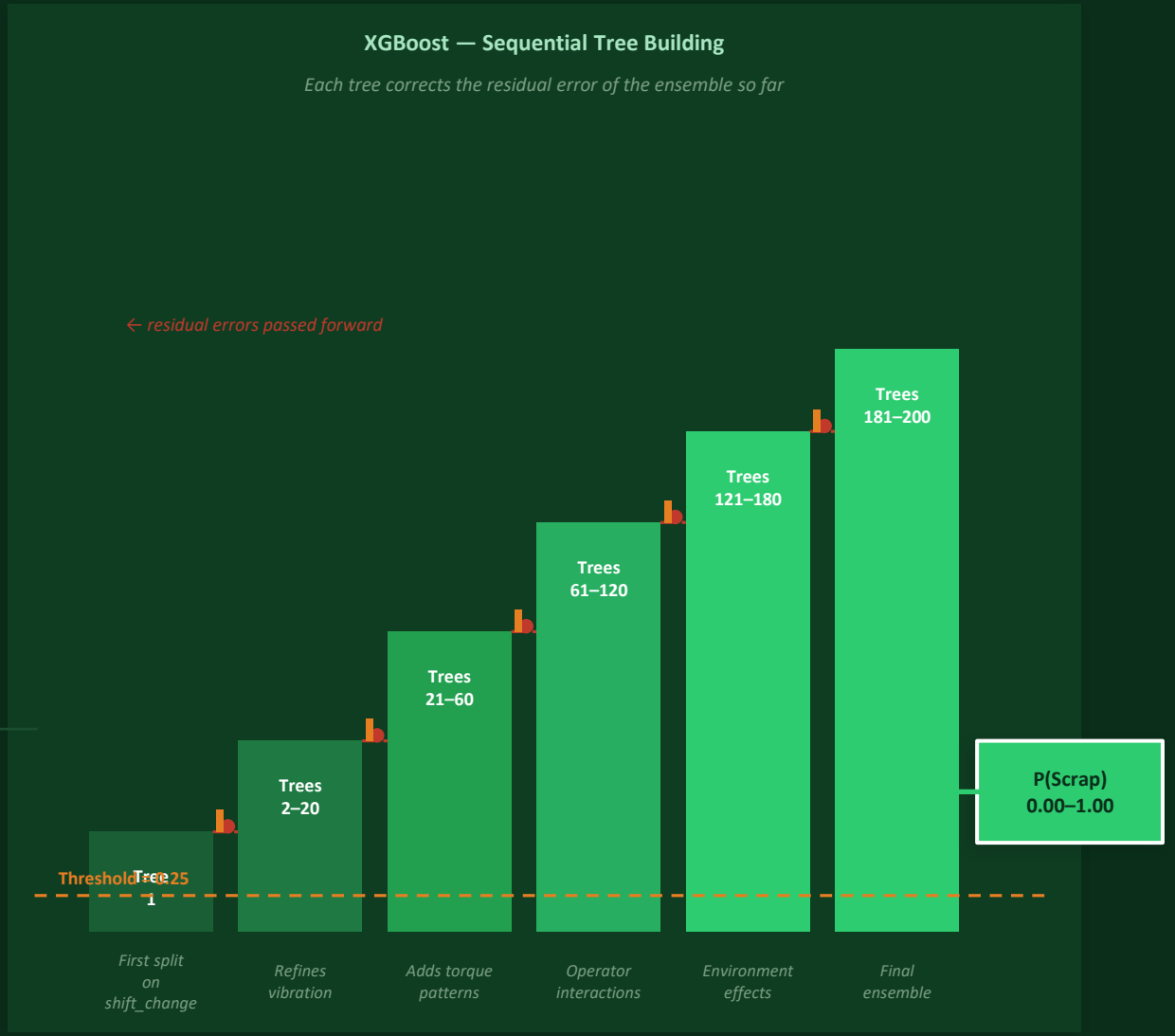
Turning Operations into Predictive Systems



The threshold that catches 69% of scrap is not a model parameter — it is a business decision about the cost of a miss.

The Real Problem

Control Charts Miss the Combination

SPC monitors variables individually. A single low-torque reading is within spec. High line vibration alone is within spec. A novice operator alone is expected. Together — under-torque + vibration spike + night shift change — they produce a scrap unit that no individual control chart predicted.

Scrap Is Total Sunk Cost

Unlike rework, scrap cannot be recovered. Materials, machine time, operator time, and downstream inspection are all gone. The mechatronics module that exits the line as scrap carries full production cost and zero recoverable value. Every percentage point of scrap rate is pure waste.

30.3% of Cycles Produce Scrap

Across 10,000 assembly cycles, nearly one in three units is scrapped. Shift changeover cycles produce 34.6% scrap vs 27.9% on normal shifts. Novice operators produce 35.3% scrap vs 29.5% for experienced ones. The patterns are real — and they interact nonlinearly.

Insight: A control chart detects one variable. XGBoost detects the combination. That difference is the entire value proposition.

The unit doesn't fail because of one bad parameter. It fails because of twelve parameters that conspired.

From SPC to Multivariate Risk Scoring

BEFORE — SPC + Manual Inspection

01 Monitor Each Variable Independently

Torque control chart. Vibration chart. Humidity log. Each watches one signal in isolation.

02 React When a Single Variable Breaches Limit

Torque falls below 1.6 Nm → operator alert. But 1.7 Nm + high vibration + novice operator → no alert.

03 Inspect After End-of-Line Scrap Decision

Quality gate confirms the unit is scrap. All production cost already committed.

04 Post-Mortem Analysis

Engineer reviews the charts — finds no single variable triggered. Root cause inconclusive. "Must be the batch."

05 Pattern Repeats Next Shift Change

No model of the interaction. No early warning. The same combination of risks produces scrap again.

VS

AFTER — XGBoost Scrap Risk Score

01 All 12 Signals Enter Simultaneously

Torque, vibration, operator experience, shift change, humidity — all evaluated jointly at every cycle.

02 200 Trees Learn the Combinations

Each tree specialises on the patterns the previous ones missed. Shift change + low torque + vibration gets its own branch.

03 P(Scrap) Predicted Before Dispatch

Model outputs a probability at the end of assembly — before the quality gate. Threshold 0.25 flags the unit.

04 Operator Receives Pre-Dispatch Alert

"This unit has 62% scrap probability — route to manual inspection before dispatch." 69% of actual scrap units caught.

05 Pattern Encoded in the Model

The shift-change + novice operator combination is permanently encoded. It will trigger on the next occurrence — automatically.

SPC detects one variable out of range. XGBoost detects a risky combination of all twelve — before the unit is completed.

The Data

Where Each Variable Lives in the Assembly Cell

Variable	Source System	Operational Context
applied_torque_nm, screwdriving_speed_rpm, motor_current_a	Torque Controller PLC	Peak torque, speed & current per fastening cycle — waveform summarised at cycle close
press_pressure_bar	Press Station PLC / Pressure Transducer	Peak pressure per press stroke — logged at each station event
operator_cycle_time_s	MES / Barcode Scanner	Timestamp delta: unit scan-in → scan-out at assembly station
relative_humidity_pct, shop_floor_temp_c	Environmental Monitoring System	Bay-level sensors logged every 5 min — joined to production records by timestamp
line_vibration_mm_s	Vibration Sensor / SCADA	RMS vibration from accelerometer on assembly frame — 1-min rolling average
operator_experience_yrs	HR System / MES	Years-of-service lookup from operator badge ID at station login
shift_change	MES / Shift Schedule	Binary flag derived from production timestamp vs shift boundary table
material_batch	ERP / Goods Receipt	Batch ID from material traceability record — linked to production order
is_scrap	MES / Quality Station	TARGET — scrap disposition at end-of-line quality gate

Dataset Profile

Records	10,000
Features	12
Train	8,000
Test	2,000
Scrap rate	30.3%
OK:Scrap	2.3:1
Shift chg scrap	34.6%
Normal shift	27.9%
Novice(0-1y)	35.3%
Senior(10+y)	29.5%

This dataset joins 5 systems on timestamp + operator badge ID + material batch ID. The scrap decision may be recorded up to 30 min after the assembly event — requiring time-alignment before the first training row can be built.

XGBoost requires no feature scaling — tree splits are rank-based, not distance-based.

Why XGBoost?

"Logistic regression finds a linear boundary. XGBoost finds the boundary the data actually has — regardless of its shape."

01 Sequential Correction — Learns From Its Own Errors

Each tree is fitted to the residuals of the entire ensemble so far — not to the raw data. The difficult cases near the decision boundary receive progressively more attention as trees are added. After 200 iterations, the ensemble has specialised coverage across the full risk landscape.

02 Interaction Effects Captured Natively

The nonlinear jump in scrap risk when vibration exceeds 3.5 mm/s — or the bimodal torque pattern where both under-torque AND over-torque increase risk — emerge automatically from the tree structure. No feature engineering required. The tree splits find these thresholds from the data.

03 No Feature Scaling Required

XGBoost splits are based on rank-ordered thresholds, not distances. A feature measured in Nm and one measured in years carry equal influence per split — making it the only model in this portfolio that does not require StandardScaler as a preprocessing step.

04 SHAP Provides Local Explainability

For each flagged unit, SHAP values answer: 'which features pushed this specific prediction toward scrap?' The quality engineer receives not just a risk score but a ranked list of contributing factors for the unit in front of them — without opening the notebook.

200 trees each thinking about all 12 variables simultaneously. That is the difference between SPC and XGBoost.

Model Results

Performance at Two Thresholds — Threshold Is a Business Decision

Default thr = 0.50

0.593	14%	47%	0.22
ROC-AUC	Recall	Precision	F1

Operational thr = 0.25

0.593	69%	34%	0.46
ROC-AUC	Recall	Precision	F1

Feature Importance (XGBoost Gain)

shift_change	0.138	Highest gain — shift boundaries create clearest tree splits
line_vibration_mm_s	0.101	Strongest continuous driver — threshold + directional effect
relative_humidity_pct	0.085	Environmental contamination — adhesive & connector sensitivity
press_pressure_bar	0.083	Press station quality signal
applied_torque_nm	0.079	Bimodal: too low AND too high both increase risk — nonlinear
operator_experience_yrs	0.075	Threshold: novice (0-1y) 35.3% vs senior 29.5% scrap

Confusion Matrix · thr=0.25 · n=2,000

TN 589	FP 806
FN 190	TP 415

Caught 415 of 605 actual scrap units

From Insight to Action

01 — Quality Engineer

Set Threshold Based on Cost Function, Not Statistics

If the assembled unit costs €500 and downstream slipped scrap risk is €2,000, the break-even precision is 500/2500 = 20%. The model achieves 34% at threshold 0.25 — above break-even. Run this calculation for your actual cost structure before choosing the operating threshold.

IMPACT

Economically justified early-warning system

02 — Production Supervisor

Route Shift-Change Cycles to Manual Inspection

Shift changeover is the single highest-gain feature. Cycles flagged during the 30-minute changeover window receive a mandatory manual inspection pass before dispatch. This targets the 34.6% scrap rate segment with zero model involvement — pure operational logic.

IMPACT

6.7pp scrap reduction on changeover cycles

03 — HR / Training

Pair Novice Operators with Mentors During Shift Changes

Novice operators (0–1yr) produce 35.3% scrap — 5.8 points above experienced staff. The effect flattens after year 2. The highest-risk combination is novice + shift change: double the structural risk factors. A buddy-system protocol during shift transitions is the lowest-cost intervention.

IMPACT

Addresses highest compound risk combination

04 — Maintenance

Add Vibration Alert at 3.5 mm/s Threshold

Line vibration is the strongest continuous process driver. The SHAP analysis and 2D risk map confirm that scrap probability rises sharply above ~3.5 mm/s. A dedicated vibration alert at this threshold — independent of SPC control limits — routes high-vibration cycles to inspection automatically.

IMPACT

Actionable threshold derived from model, not intuition

05 — ML Practitioner

Enrich material_batch With Incoming Inspection Metadata

The model encodes batch as a numerical ID 1–50 — it cannot distinguish a critical batch with a supplier deviation from a nominal one. Adding incoming inspection pass/fail, supplier certificate deviation flag, and certificate date as features would likely improve precision meaningfully at the next retraining cycle.

IMPACT

Highest-ROI feature engineering for iteration 2

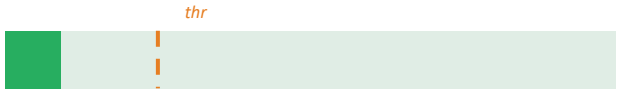
The model catches what charts miss. The threshold catches what statistics ignore. The mentor catches what data will never see.

What-If Simulator

Same line, same shift — three different risk combinations. The model assigns each unit a scrap probability before dispatch.

Scenario A — Controlled Conditions

Torque	1.85 Nm
Vibration	1.0 mm/s
Operator Exp	5 years
Shift Change	No
Humidity	40%



P(Scrap) = 9.1%

✅ **OK to proceed**

Scenario B — All Risk Factors

Torque	1.4 Nm ← low
Vibration	4.8 mm/s ← high
Operator Exp	0 years ← novice
Shift Change	YES ← changeover
Humidity	70%

P(Scrap) = 92.8%

⚠ **Flag as SCRAP RISK**

Scenario C — Process Fixed

Torque	1.85 Nm ✓
Vibration	1.5 mm/s ✓
Operator Exp	0 years ← remains
Shift Change	YES ← remains
Humidity	70%

P(Scrap) = 62.0%

⚠ **Still flagged**

B→C: Fixing torque + vibration drops risk from 92.8% to 62.0% — a 31pp reduction. Human factors (novice + shift change) carry residual risk that process adjustment alone cannot correct.

The Supervised Learning Journey

Twelve projects. Two families. One question — what does the data need?

CLASSIFICATION — Projects 01–06

- 01 Logistic Regression
- 02 Naive Bayes
- 03 KNN
- 04 SVM
- 05 Decision Tree
- 06 Random Forest

REGRESSION — Projects 07–11

- 07 Linear Regression
- 08 Multiple Regression
- 09 Lasso L1
- 10 Ridge L2
- 11 Elastic Net

Can do both

12

XGBoost

Classification

& Regression

Classification models

Projects 01–06: from Logistic Regression to Random Forest — boundaries from simple to complex.

Regression models

Projects 07–11: from Linear OLS to ElasticNet — additive physics to adaptive regularisation.

12 XGBoost closes both

The same XGBoost API handles classification and regression. It is not the end of supervised learning — it is the pivot to the next chapter.

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